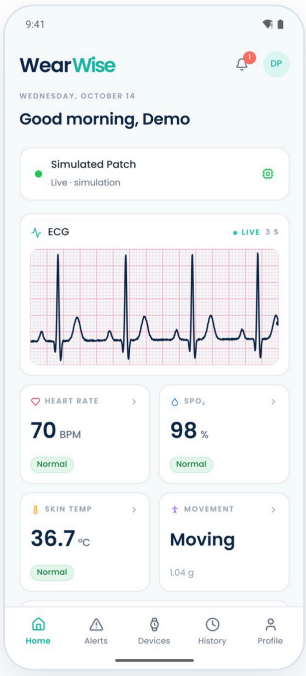
WearWise

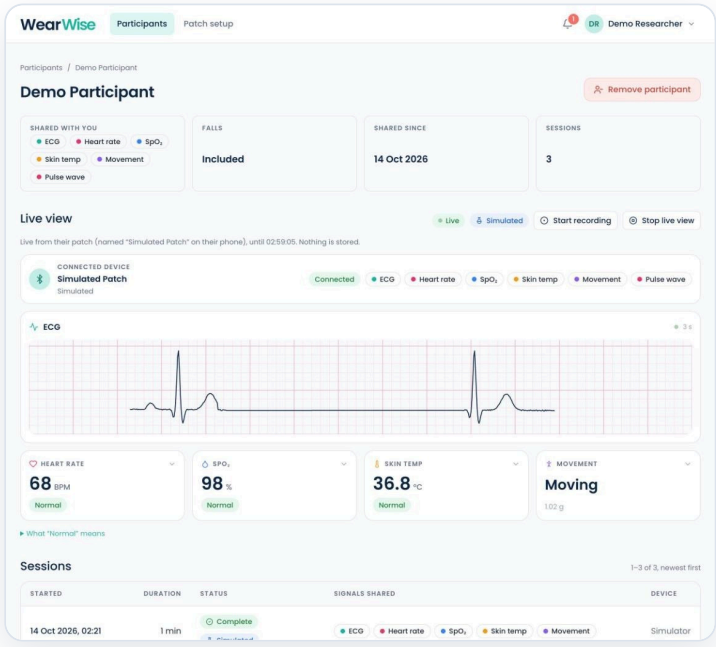
USER GUIDE

The WearWise app and web dashboard

How to use every screen of the WearWise phone app and the researcher dashboard, step by step, with screenshots.







Handover release v1.0.0 · app 0.1.0 · firmware 0.9.6

October 2026 · WearWise Project Team

Zakiul Atheef · Hari Bethelli · Siri Bethelli · Thevindu Kodagoda Munasinghage

Research prototype.
Not a medical device.

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PART 1

Getting to know WearWise

What WearWise is, who uses which part, and the words this guide uses.

1 What WearWise is

WearWise is a **research prototype** for collecting wearable sensor data with the participant in control. It has three parts that work together:

1

The patch

A small sensor board worn on the body. It measures heart rate, blood oxygen (SpO_2), pulse wave, skin temperature, movement and ECG, and sends the readings to the phone over Bluetooth.

2

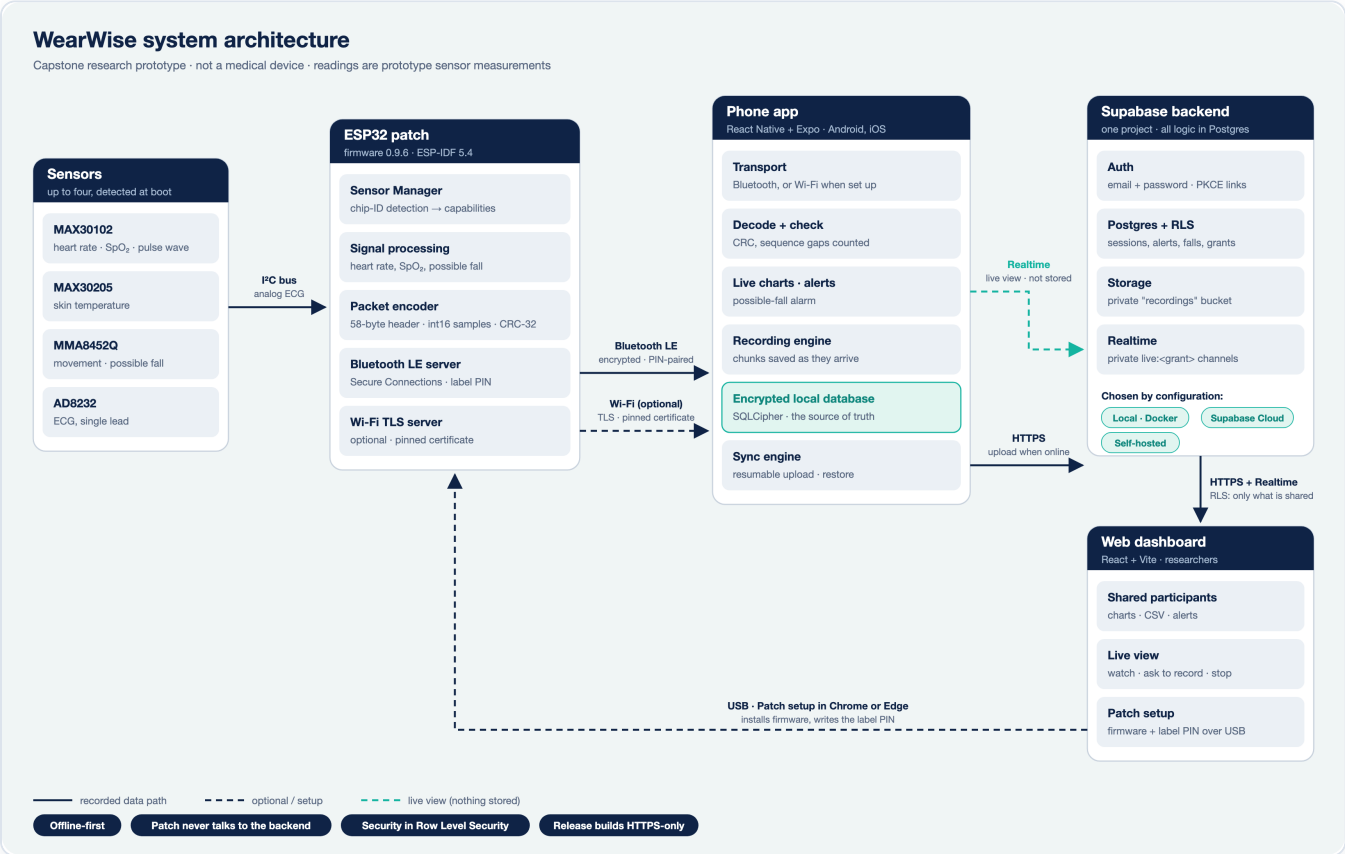
The phone app

Shows the readings live, records sessions, warns about a possible fall, and backs recordings up. The participant decides who can see what.

3

The web dashboard

Where a researcher or clinician sees the data a participant chose to share: sessions, charts, alerts and live view. It also installs firmware on patches.



How the parts connect. The patch talks only to the phone. The phone keeps every recording in an encrypted database on the phone and backs it up to the project's backend. The dashboard shows only what the participant shares.

Important: WearWise is not a medical device

Its readings are prototype sensor measurements and are not clinically validated. "Possible fall detected" means the movement looked like a fall. It can miss falls and it can raise false alarms. No emergency service or contact is ever alerted. Don't use WearWise to diagnose, treat or monitor any medical condition. If you feel unwell, contact a doctor or emergency services directly.

2 Who uses what

Person	Uses	Can do
Participant	The phone app (and, optionally, the dashboard's *My data* pages)	Wear the patch, see live readings, record sessions, answer fall alarms, and decide who sees which signals. Can stop sharing at any time.
Researcher or clinician	The web dashboard	See participants who shared with them, open sessions and charts, download CSV files, receive alerts, watch live view while a participant allows it, and set up patches.
Administrator	The backend (see the Setup Guide)	Creates researcher and clinician accounts. Participants create their own accounts in the app.

3 Words used in this guide

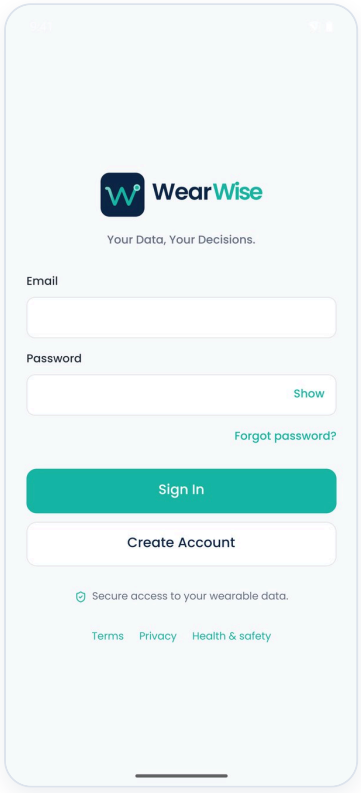
Patch / sensor	The WearWise sensor board. Its Bluetooth name looks like `WearWise-6280`.
Simulated patch	A pretend patch built into the app. It produces realistic test data, so you can try everything without hardware. Simulated data is always labelled **Simulated**.
Signal	One kind of measurement: ECG, heart rate, SpO_2, skin temperature, movement or pulse wave.
Session / recording	A stretch of time you chose to save, with the signals you picked.
Backed up	A copy of the session has been uploaded to the project's backend (the "cloud").
Share code	A one-time code, like `AWWXK-59EF5`, that a participant gives a researcher so they can see chosen signals.
Live view	A temporary mode in which a researcher can watch the readings as they arrive. Nothing is saved.

PART 2

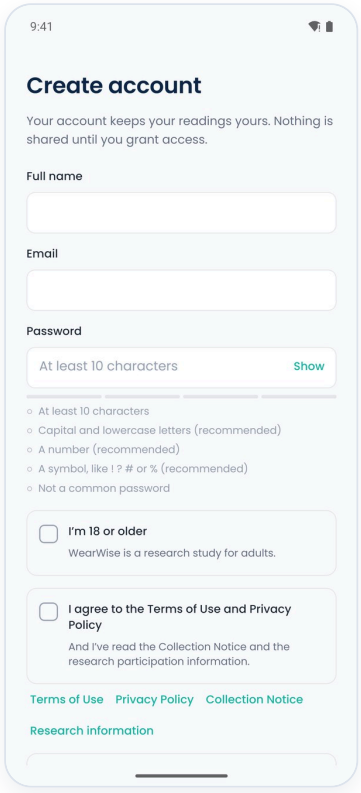
The phone app

For participants. Screens are shown from an Android phone. iPhone looks the same.

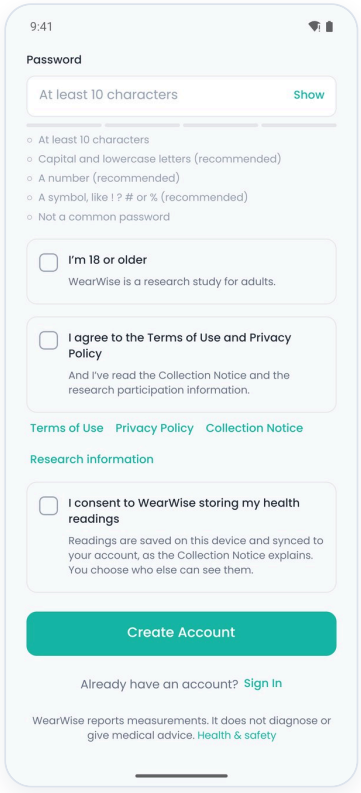
4 Signing in and creating an account



Sign in. Email and password, or **Create Account**.



Create account. Name, email and a password.



Agreements. Tick all three boxes to continue.

Create an account

1. Open WearWise and tap **Create Account**.
2. Enter your **full name** and **email address**.
3. Choose a **password of at least 10 characters**. The checklist under the box turns green as you go. Capital letters, a number and a symbol are recommended, and very common passwords are refused.
4. Tick **I'm 18 or older**. WearWise is a research study for adults.
5. Tick **I agree to the Terms of Use and Privacy Policy**. The links under the box open each document.
6. Tick **I consent to WearWise storing my health readings**, then tap **Create Account**.

Depending on how the study is set up, you may need to confirm your email address first: open the email WearWise sends and tap its link.

Forgot your password?

On the sign-in screen tap **Forgot password?**, enter your email and follow the link in the email you receive. The link works once and expires after an hour.

5 Finding your way around

The bar at the bottom of the screen has five tabs. You can reach every feature from one of them.

Tab	What it's for
Home	Live readings from your patch, recording, recent alerts and this week's summary.
Alerts	Possible falls and readings that stayed outside the usual range.
Devices	Pair, connect, rename or forget a patch; use the simulation; Wi-Fi streaming; automatic backup.
History	Every session you recorded, with charts.
Profile	Your details, privacy and sharing, backup, password, notifications, documents and sign-out.

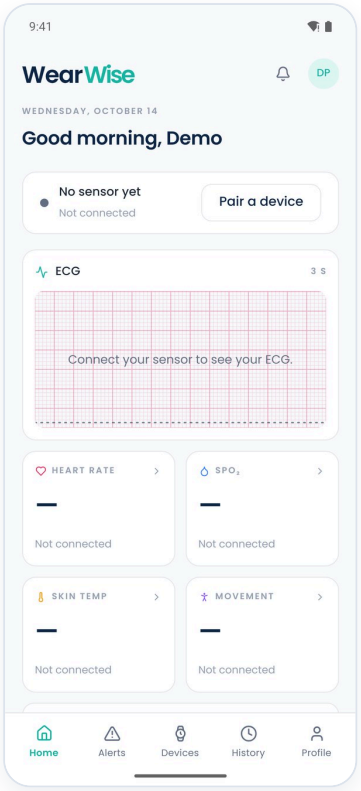
The **bell** at the top of Home opens your notifications. The circle with your initials opens your profile.

6 The Home screen

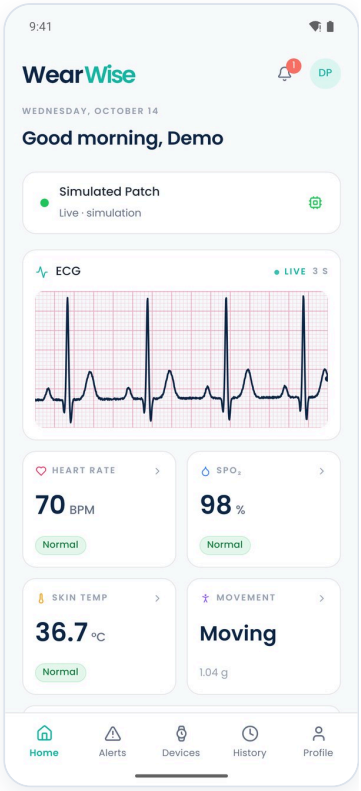
Home is where you see your readings as they arrive. From top to bottom:

- **Sensor card.** Which patch is connected and how. With no patch it says *No sensor yet* and offers **Pair a device**.
- **ECG strip.** The last 3 seconds of your heart's electrical signal. LIVE shows it is updating.
- **Reading tiles.** Heart rate (BPM), SpO_2 (%), skin temperature (°C) and movement (*Still*, *Moving* or *Active*). Each tile says Normal, Below normal or High.
- **What "Normal" means.** Explains the reference ranges (table below).
- **Record.** Starts a recording (section 8).
- **Alerts** and **This week.** Your latest alerts, then minutes recorded, sessions and how many are backed up. **Sync** backs up now.

A tile shows a dash (—) and a short reason when there's no reading, for example *Not connected* or *Finger on the sensor*. Signals your patch doesn't have say so.



No patch connected yet.



Live readings from a patch.

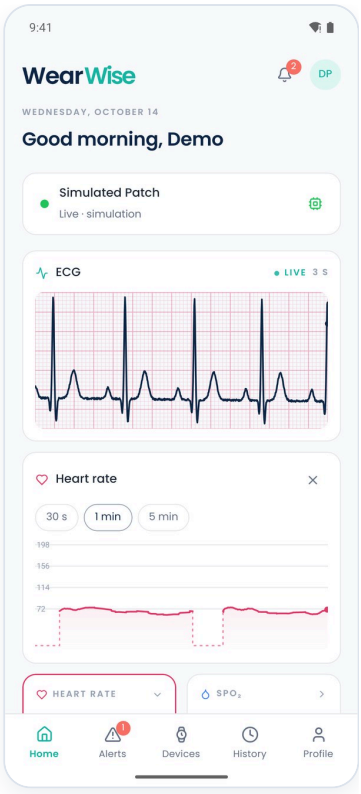
See more detail

Tap a tile (for example **Heart rate**) to open its chart under the ECG. Choose **30 s**, **1 min** or **5 min** to change how much history it shows. Tap **×** to close it.

The reference ranges

Signal	"Normal"	An alert opens when…
Heart rate	60–100 BPM	below 60 or above 100 for 15 s
SpO_2	95 % or more	below 95 % for 15 s
Skin temp	31–37.5 °C	above 37.5 °C for 60 s

These ranges are general reference values for adults at rest, used to label prototype measurements. They are not a diagnosis. Skin temperature is lower than core body temperature.

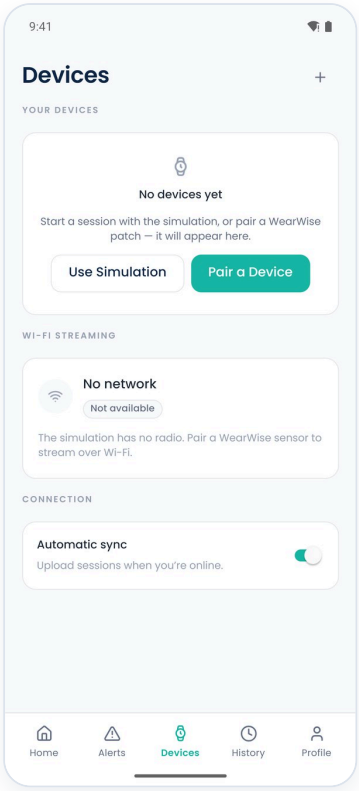


Heart rate opened, showing the last minute.

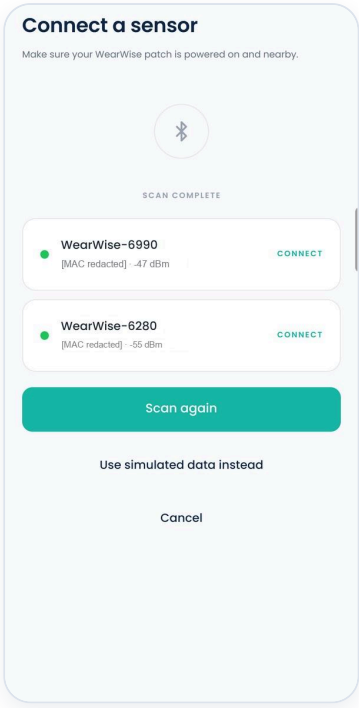
7 Connecting your patch

Open the **Devices** tab. The first time, it says *No devices yet* and offers two buttons:

- **Pair a Device**: connect a real WearWise patch over Bluetooth.
- **Use Simulation**: use the built-in simulated patch. Use this to try the app, or on an emulator, which has no Bluetooth.



Devices, before any patch.



Two patches found while scanning (real phone).

Pair a real patch

1. Switch the patch on and keep it near the phone. If the phone asks for **Nearby devices** or Bluetooth permission, allow it.
2. Tap **Pair a Device**. The app scans and lists every WearWise patch nearby by name, such as `WearWise-6280`, with its signal strength.
3. Tap **CONNECT** next to **your** patch. The name is on the patch's label.
4. When the phone asks for a PIN, type the **6-digit PIN printed on the patch's label**. Android may suggest "0000 or 1234": ignore that and use the label PIN.

> **Pairing refused?**
>
> A patch accepts a new phone only when it has never been paired, or for **60 seconds after its BOOT button is pressed**. Three wrong attempts lock pairing until BOOT is pressed. In the phone's Bluetooth settings, remove any old WearWise entry, press BOOT on the patch, then pair again.

Once connected

Your patch appears under **Your devices** with a Connected badge and three buttons:

- **Disconnect** stops the connection. It changes to **Reconnect**.
- **Rename** gives the patch a name you'll recognise.
- **Forget** removes it from the app. Its past recordings stay in History.

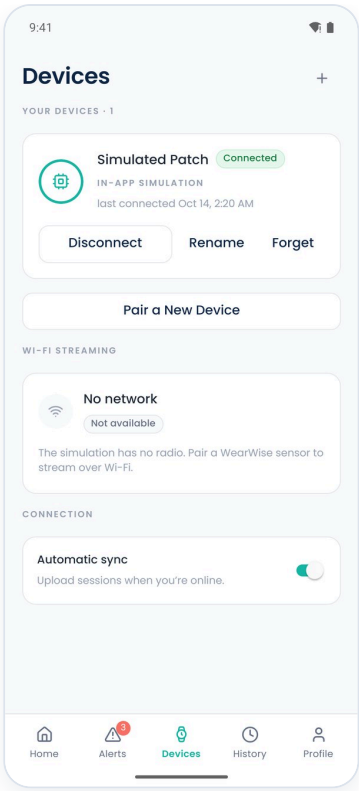
If the connection drops, the app tries again on its own and continues the same recording.

Wi-Fi streaming (optional)

A real patch can also stream over Wi-Fi when the phone and patch are on the same network. Use **Set up Wi-Fi** on the Wi-Fi card. The patch supports **2.4 GHz networks only**, and Bluetooth stays connected as a fallback. The simulation has no radio, so the card says *Not available*.

Automatic sync

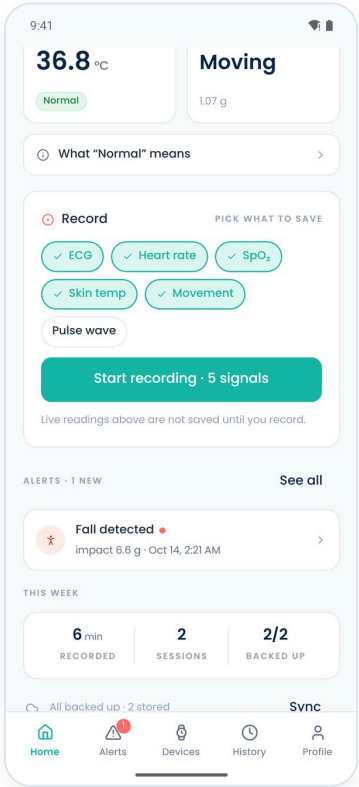
Leave **Automatic sync** on to upload sessions whenever you're online.



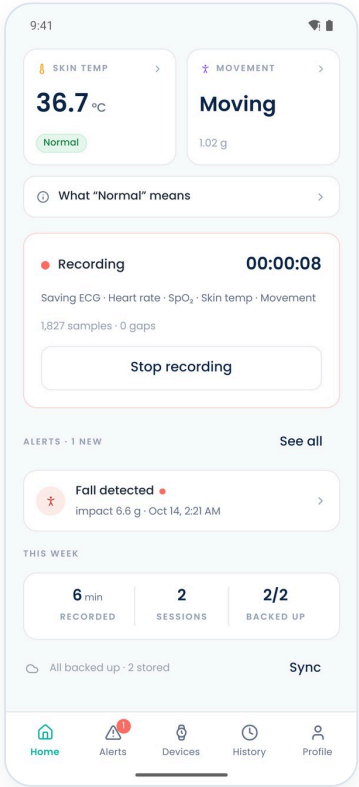
A connected patch (here the simulation).

8 Recording a session

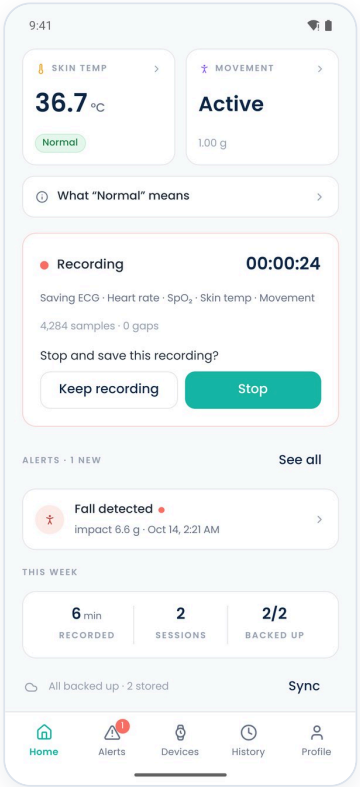
Live readings on Home are **not saved** until you record. A recording saves the signals you pick, on your phone first, then backs them up.



1. Pick what to save, then start.



2. Recording. Timer, signals, samples and gaps.



3. Stop and save. Confirm to finish.

1. Connect your patch (section 7) and scroll down Home to the **Record** card.
2. Under *Pick what to save*, tap the signals to include. Ticked chips are saved. Tap a chip again to leave it out.
3. Tap **Start recording · N signals**. A red dot and a timer show it is recording. *Samples* counts the readings saved and *gaps* counts any interruptions.
4. To finish, tap **Stop recording**, then **Stop** to confirm. **Keep recording** carries on.

The finished session appears in **History**. It's uploaded when you're online, and Home's *This week* card shows *Backed up 2/2* when everything is safe.

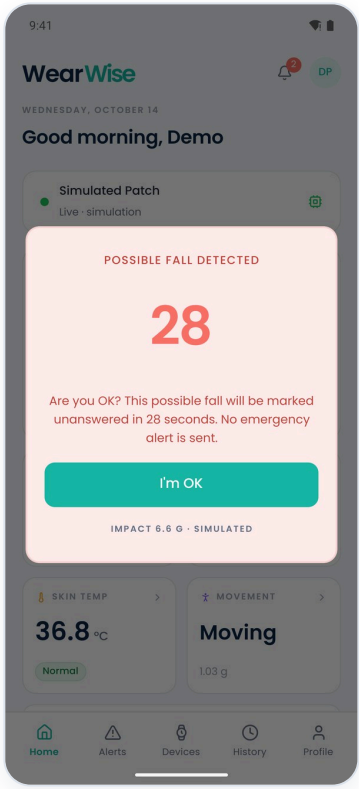
> **Offline is fine**
>
> WearWise records normally without internet. Sessions wait as "not backed up" and upload by themselves once you're online again. To start it yourself, use **Sync** on Home or **Back up now** in **Profile → Privacy & access**.

9 When a possible fall is detected

If the movement sensor sees a pattern that looks like a fall (a short drop, a hard impact, then stillness), the app shows a full-screen **POSSIBLE FALL DETECTED** card and sounds an alarm, even when the phone is on silent.

1. If you're fine, tap **I'm OK**. The alarm stops and the event is saved as *Marked OK*.
2. If nobody answers within **30 seconds**, the event is saved as *unanswered*.

The card shows the strength of the impact in *g*, and SIMULATED when it comes from the simulation.



The possible-fall alarm with its 30-second countdown.

> **No one is called**
>
> WearWise does not contact emergency services, family or anyone else. A researcher you share falls with sees the event on the dashboard, but only when they're looking. Fall detection is a simple prototype rule and can be wrong in both directions.

> **Using the simulation?**
>
> The simulated patch includes a pretend fall about 12 seconds after it starts streaming (on connecting, and again when a recording starts or stops), then every 3 minutes, so you can practise. Just tap **I'm OK**.

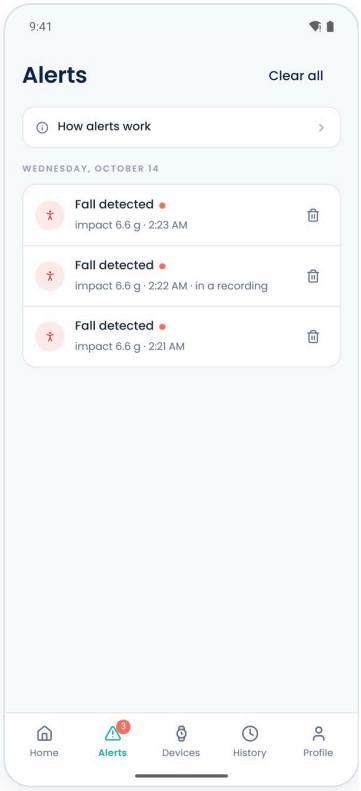
10 Alerts

The **Alerts** tab lists, newest first:

- **Fall detected**: every possible fall, with the impact, the time and whether it happened during a recording.
- **High / low readings**: a heart rate, SpO_2 or skin temperature that stayed outside its range for the time in the table in section 6. One alert covers the whole time it stayed out of range and shows its peak.

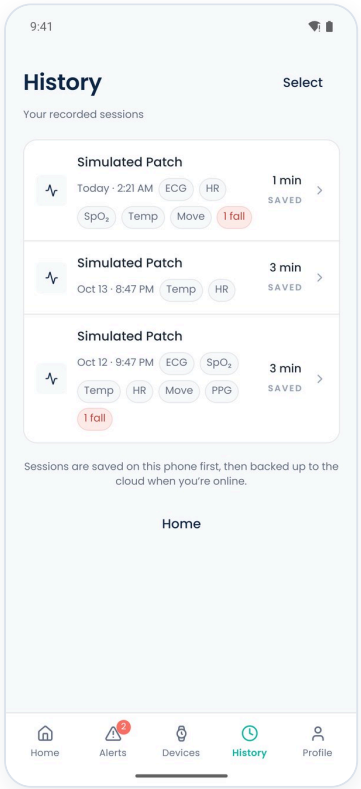
A red dot marks alerts you haven't seen. The number on the tab icon counts them. Tap the bin icon to delete one alert, or **Clear all** to delete them all. **How alerts work** explains the rules in the app.

In **Profile → Notifications** you can also get alerts as phone notifications while WearWise is running, and choose whether a possible fall sounds an alarm.



Three possible falls from the simulation.

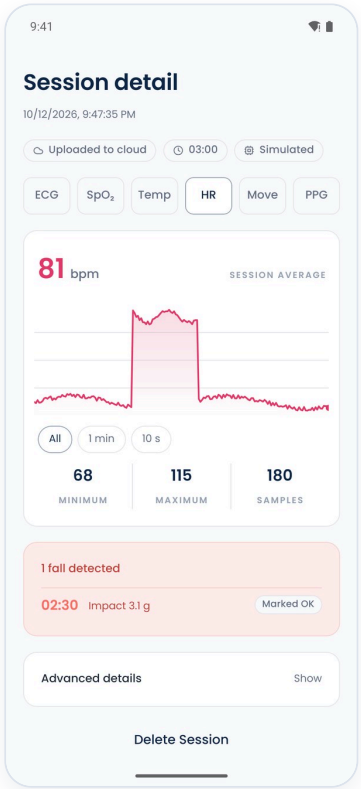
11 History and session details



History. One card per session.



Session detail. ECG for the whole session.



Heart rate tab of the same session.

History lists every session on this phone: the patch, date and time, signals saved, falls (if any) and length. Sessions are saved on the phone first, then backed up. **Select** lets you pick several sessions at once.

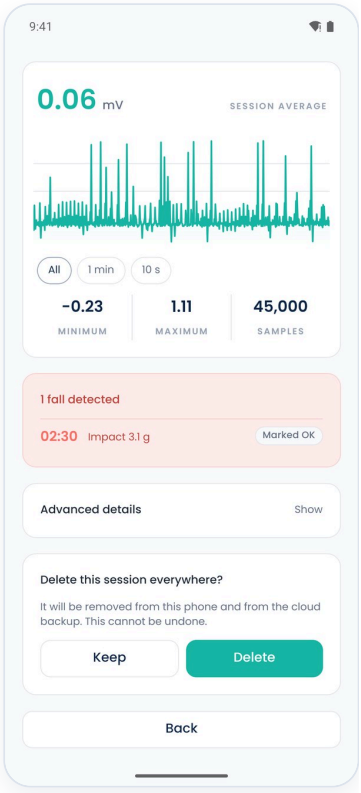
Tap a session to open it:

- Badges at the top show whether it's *Uploaded to cloud*, how long it is, and whether it's simulated.
- Tabs switch between signals: **ECG, SpO_2, Temp, HR, Move, PPG** (pulse wave).
- The large number is the session average. Below the chart, **All**, **1 min** or **10 s** zooms the chart. *Minimum*, *Maximum* and *Samples* summarise the signal.
- Falls during the session are listed with the time into the session, the impact and the answer (for example *Marked OK*).
- **Advanced details** shows technical information about the recording.

Deleting a session

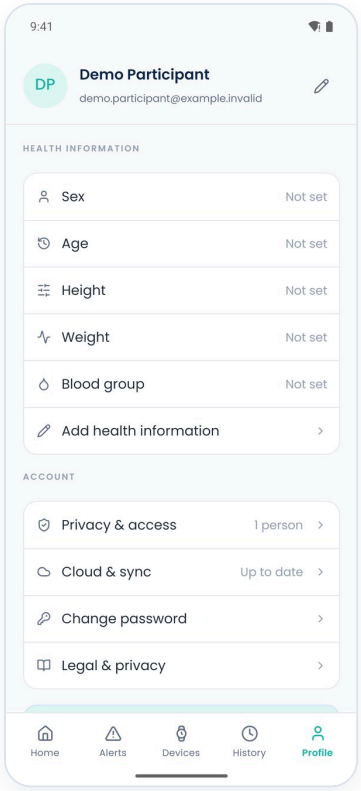
At the bottom of a session, tap **Delete Session**. WearWise asks first, because a deleted session is removed **from this phone and from the cloud backup**, and can't be undone. Tap **Delete** to confirm or **Keep** to cancel.

A researcher you shared with also stops seeing a deleted session.

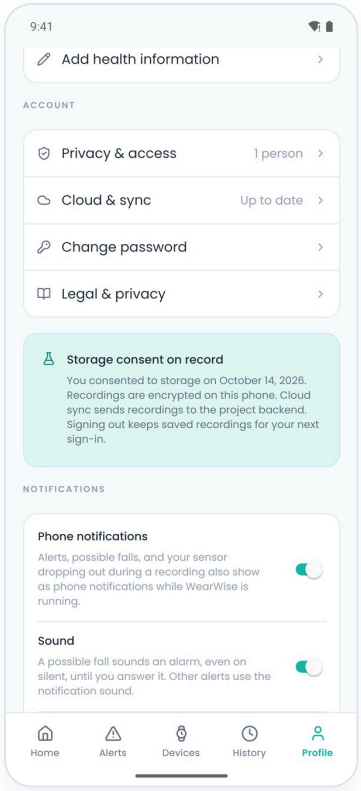


Deleting asks for confirmation.

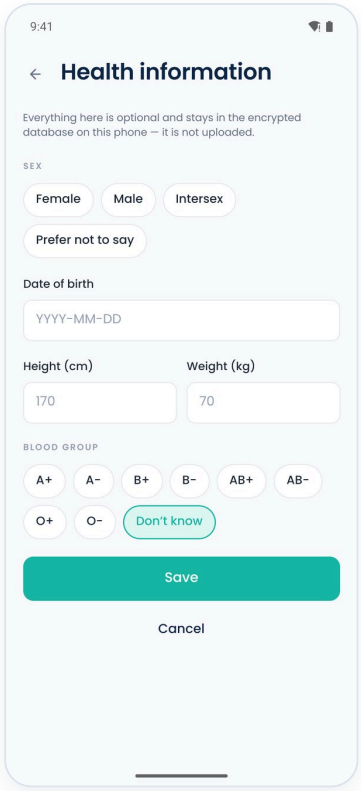
12 Profile and settings



Profile. Your details and health information.



Account and notifications, further down.



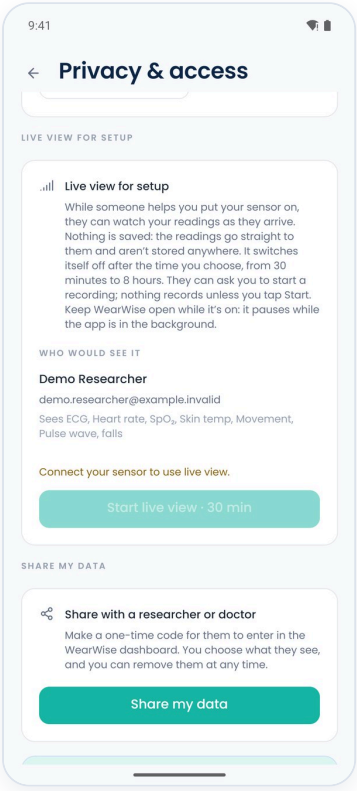
Health information. Optional; stays on the phone.

Name and picture	Tap the pencil next to your name to change your name or profile picture.
Health information	Sex, date of birth, height, weight and blood group. All optional. It is stored only in the encrypted database on your phone and **is not uploaded**.
Privacy & access	Who can see your data, sharing codes, live view and backup status (sections 13 and 14).
Cloud & sync	Whether everything is backed up (*Up to date*).
Change password	Enter your current password, then a new one (10+ characters).
Legal & privacy	The Terms of Use, Privacy Policy, Collection Notice, research information and health & safety notes, and at the end the team who made WearWise and its licence.
Storage consent on record	The date you agreed to WearWise storing your readings.
Notifications	**Phone notifications**: show alerts and patch drop-outs as notifications while WearWise is running. **Sound**: a possible fall sounds an alarm, even on silent, until you answer it.
Appearance	Turn on the dark appearance if you prefer it.

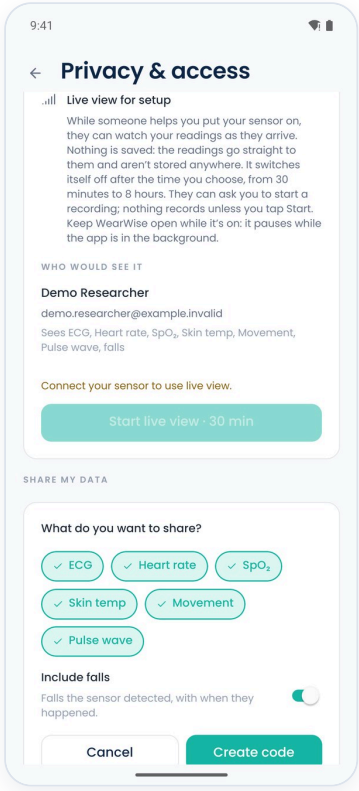
Export all my data / Request account deletion	Opens an email to the study's privacy contact so they can act on your request (or the privacy policy, if no contact has been set up).
Sign out	Asks first. Your saved recordings stay on this phone for your next sign-in.

13 Sharing your data with a researcher

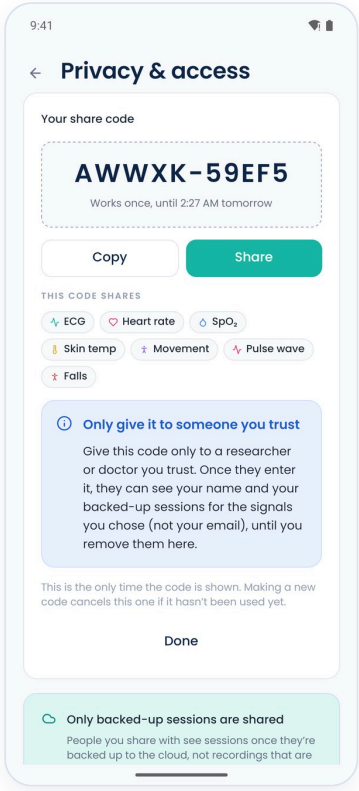
Nobody can see your recordings until you share them. You decide **who**, **which signals** and **whether falls are included**, and you can change or end it at any time. Sharing is done with a one-time **share code**.



Privacy & access → Share my data.



Choose the signals and whether to include falls.



Your code. It works once, for 24 hours.

1. Go to **Profile → Privacy & access** and scroll to **Share my data**.
2. Tap **Share my data**. Under *What do you want to share?*, tick the signals the researcher may see. Turn **Include falls** on or off.
3. Tap **Create code**. Your code appears, for example `AWWXK-59EF5`. It works **once** and expires after **24 hours**.
4. Tap **Copy** or **Share** and give the code to the researcher. They enter it on the dashboard (section 17). Then tap **Done**.

> **Only give the code to someone you trust**
>
> Once they enter it, they can see your name and your backed-up sessions for the signals you chose (not your email), until you remove them. The code is shown only once. Making a new code cancels an unused one.

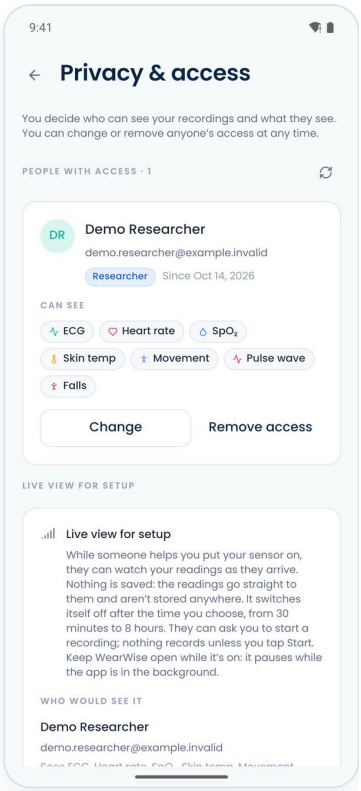
See and change who has access

People with access lists everyone you share with: their name, email, role, since when, and what they *can see*.

- **Change** picks different signals for that person.
- **Remove access** ends their access straight away.

While a code is waiting to be used, a card says *A share code is waiting to be used*, with **Make a new code** and **Cancel code**.

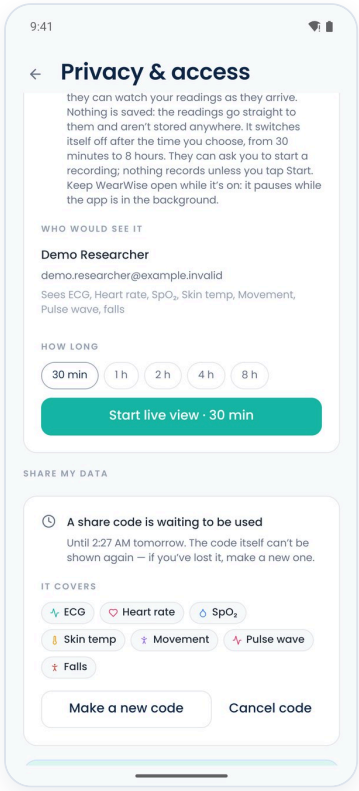
> **Only backed-up sessions are shared**
>
> People you share with see sessions once they're backed up to the cloud, not recordings that are only on your phone.



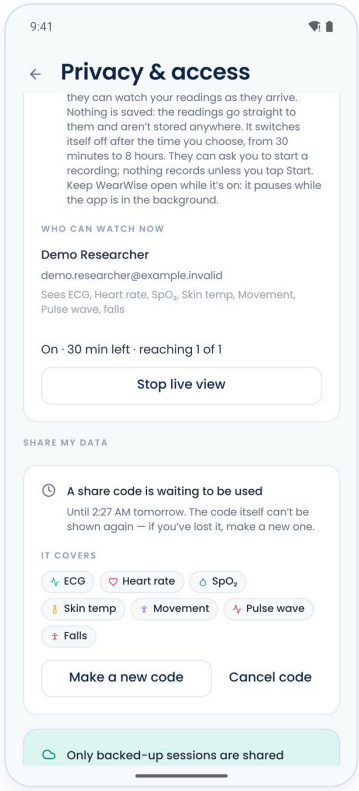
One researcher can see seven signals.

14 Live view (for setting up the patch)

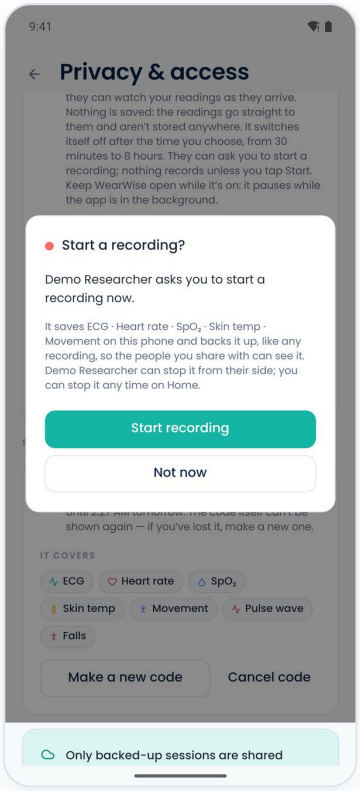
Live view lets a researcher you already share with watch your readings **as they arrive**, for example while they help you put the patch on. **Nothing is saved**: readings go straight to them and aren't stored anywhere. It switches itself off after the time you choose.



Choose how long, then start.



Live view on. Time left and Stop.



A recording request from the researcher.

1. Connect your patch. Until you do, the button says *Connect your sensor to use live view*.
2. In **Profile → Privacy & access**, find **Live view for setup**. Under *Who would see it* you can check who will be watching.
3. Under *How long*, choose **30 min**, **1 h**, **2 h**, **4 h** or **8 h**, then tap **Start live view**.
4. Keep WearWise open. Live view pauses while the app is in the background. Home shows a banner with the time left.
5. Tap **Stop live view** at any time. The researcher can also stop it from their side.

Recording requests. During live view the researcher can ask you to start a recording. A card explains what would be saved. Tap **Start recording** to agree or **Not now** to decline. Nothing records unless you tap Start, and you can stop the recording on Home at any time.

PART 3

The web dashboard

For researchers and clinicians, in any modern browser on a computer. (Patch setup needs Chrome or Edge.)

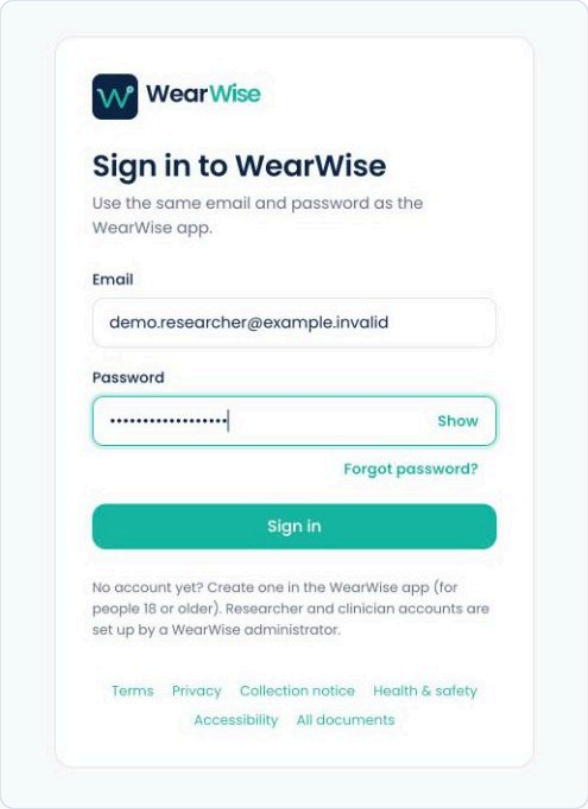
15 Signing in

Open the dashboard address your project gave you and sign in with your email and password.

- **Researcher and clinician accounts** are set up by a WearWise administrator.
- **Participants** can sign in with the same email and password as the app. They see their own data (section 22).

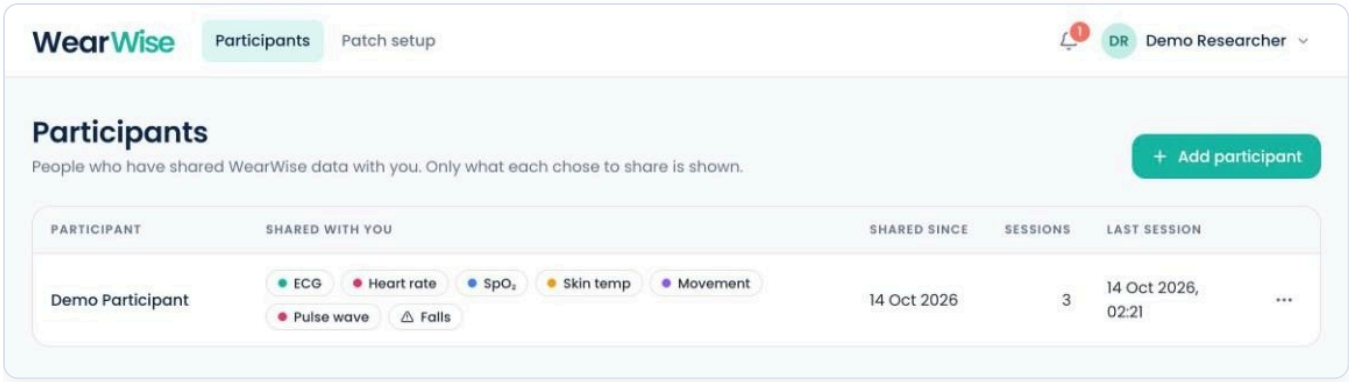
Forgot password? Under the password box, click **Forgot password?**, enter your email and click **Send reset link**. The link in the email works once and expires after an hour.

The links at the bottom of every page open the Terms, Privacy Policy, Collection Notice, Health & safety and Accessibility documents. **© 2026 WearWise Project Team** opens the team credits.



The dashboard sign-in page.

16 Participants



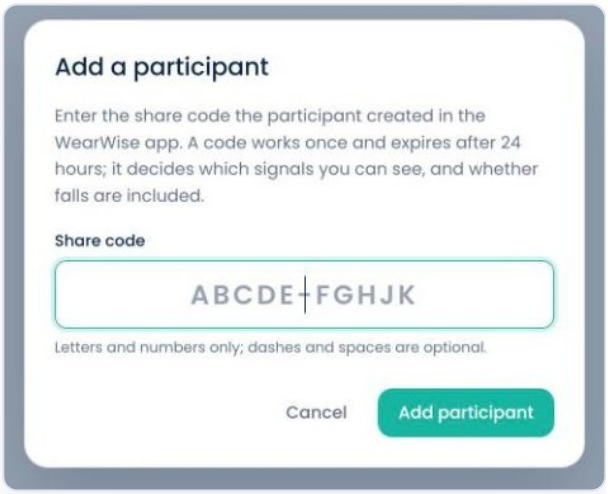
Participants. Everyone who has shared data with you, what they share, since when, how many sessions and the latest one.

After signing in you land on **Participants**. You see only people who gave you a share code, and only the signals each of them chose. A `Falls` chip means falls are included. Click a name to open that participant. The ⋯ menu at the end of a row has **Open** and **Remove participant**, which ends your access.

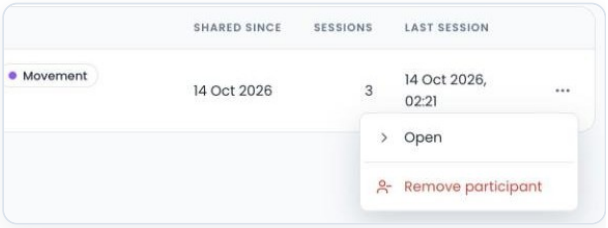
17 Adding a participant with a share code

1. Ask the participant to create a share code in their app (section 13).
2. On **Participants**, click **+ Add participant**.
3. Type the code. Capitals, dashes and spaces don't matter. Click **Add participant**.
4. The participant appears in the list straight away, with the signals they chose.

> **"Invalid or expired code"?**
>
> A code works **once**, lasts **24 hours**, and a newer code replaces an unused one. Ask for a new code. After 10 wrong tries in an hour, adding is paused for that hour.

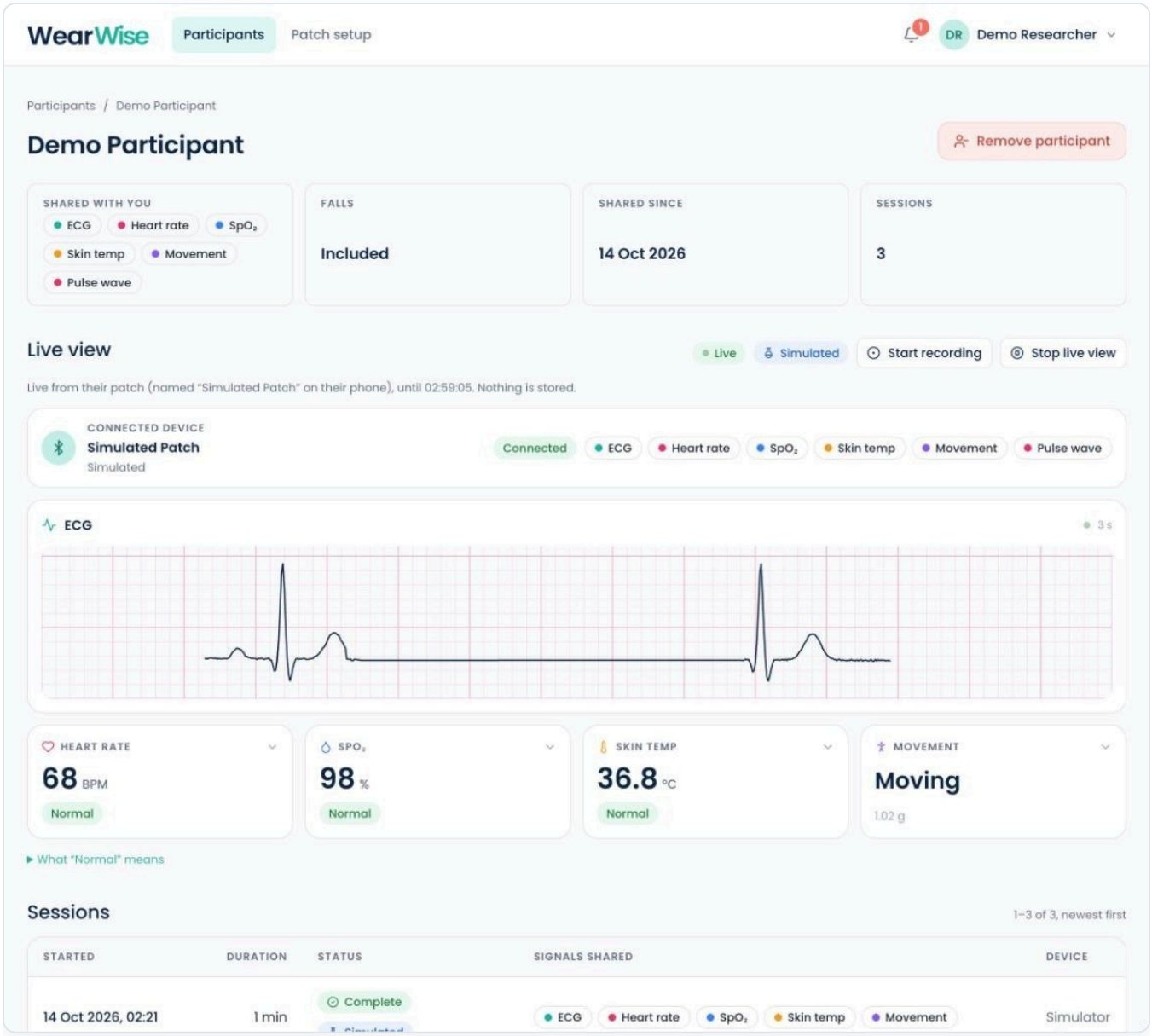


Enter the participant's share code.



The row menu: Open or Remove participant.

18 A participant's page and live view



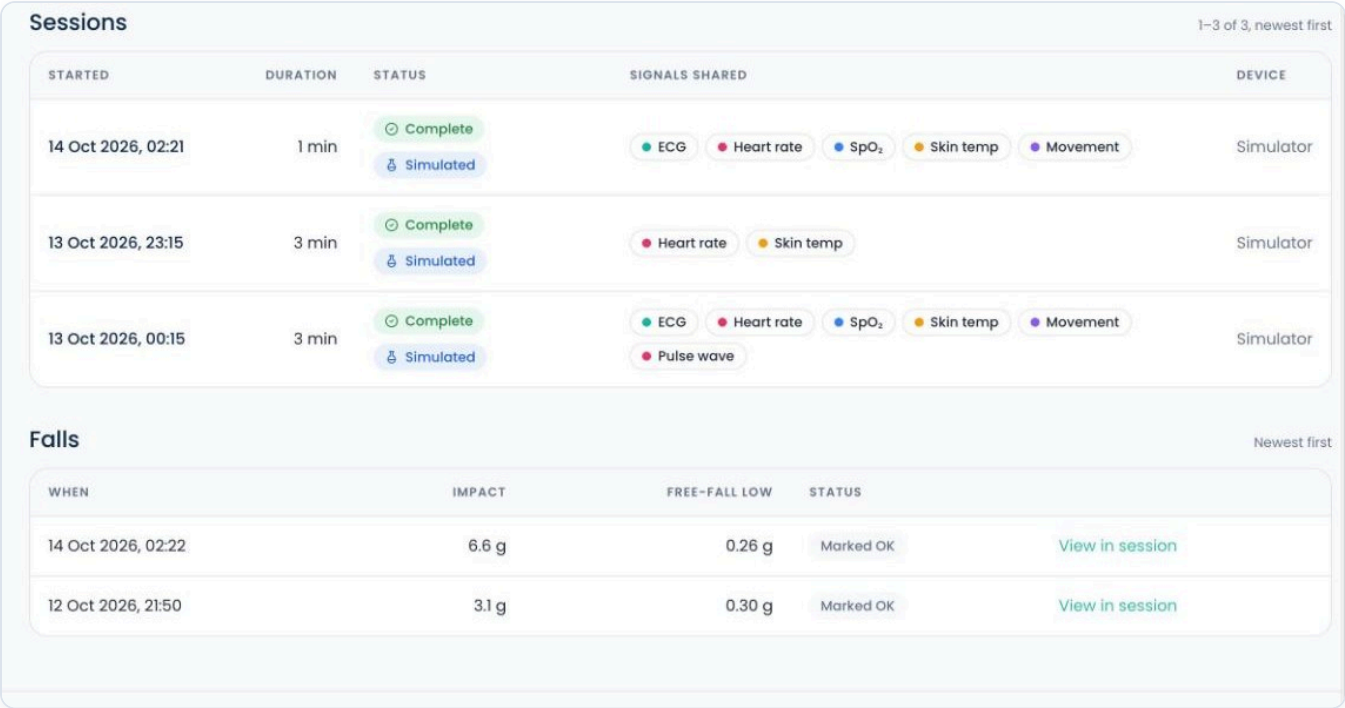
Participant page with live view on. The summary (signals shared, falls, since when, sessions), then the live ECG and reading tiles while the participant allows it.

The top of the page summarises the sharing. When the participant has **live view** on, the **Live view** card shows their connected patch, the live ECG and the reading tiles, labelled `Live` (and `Simulated` for simulated data). The text under the title says until when, and that *nothing is stored*.

- **Start recording** sends the participant a request. Nothing records unless they tap Start on their phone. While it records, **Stop recording** ends it. The session appears under **Sessions** once it's backed up.
- **Stop live view** ends live view for both of you.
- **Remove participant** (top right) ends your access.

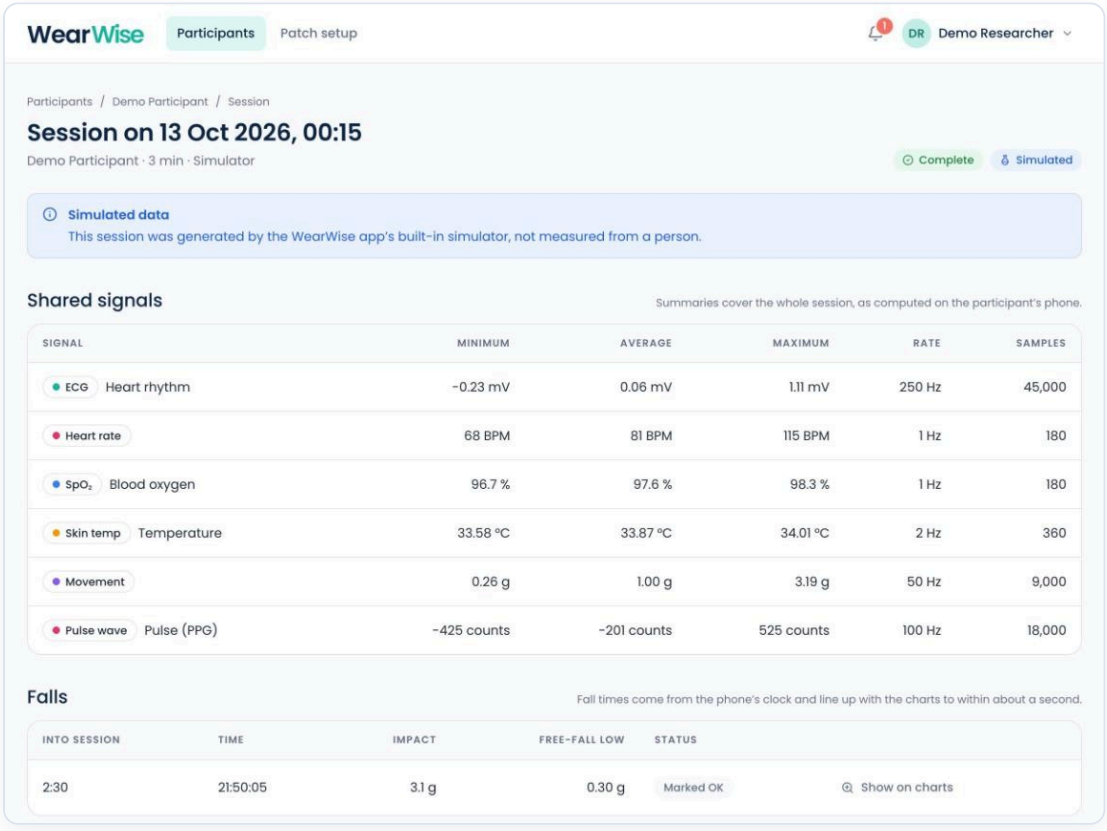
When live view is off, the card says the readings will appear once the participant switches on *Live view for setup* in their app. If their app goes to the background, it shows `Paused` until they open it again.

19 Sessions, falls and charts

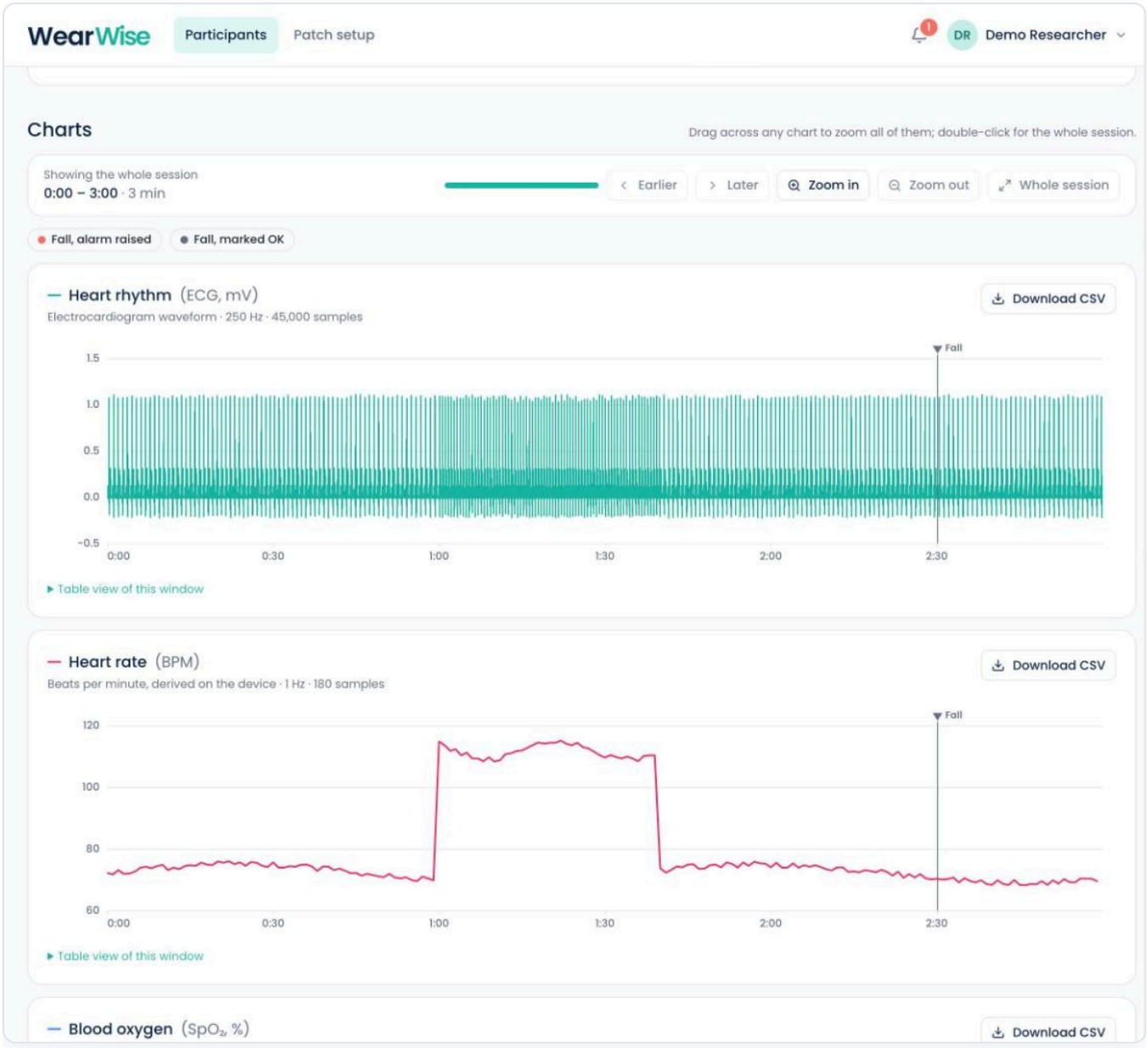


Sessions and Falls, further down the participant page: each session's start, length, status and signals, and every fall with its impact and answer.

Further down the participant page, **Sessions** lists each backed-up session with its start time, length, status (Complete , Simulated), the signals shared and the device. **Falls** lists possible falls with the impact, the lowest reading during the drop (*free-fall low*) and the participant's answer. Click a session, or **View in session**, to open it.



Session detail. A banner marks simulated data. The table summarises each shared signal over the whole session.



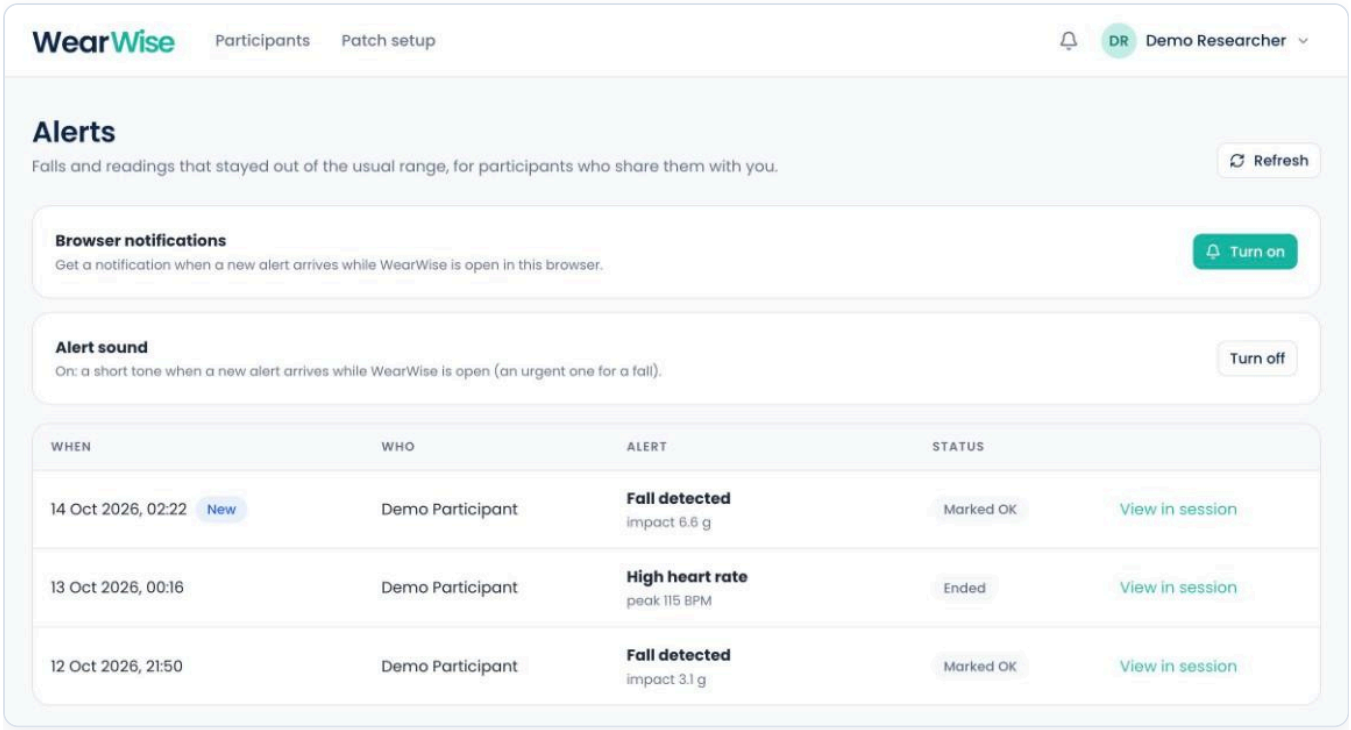
Charts. All charts share one time axis. The vertical line marks the possible fall.

Using the charts

- **Zoom:** drag across any chart to zoom all of them to that time. Double-click to see the whole session again. The buttons do the same: **Earlier**, **Later**, **Zoom in**, **Zoom out**, **Whole session**.
- **Falls** appear as vertical markers. **Show on charts** in the Falls table jumps to one.
- **Download CSV** on each chart saves that signal as a spreadsheet file.
- **Table view of this window** shows the numbers behind what's on screen.

Summaries are computed on the participant's phone over the whole session. Fall times come from the phone's clock and line up with the charts to within about a second.

20 Alerts



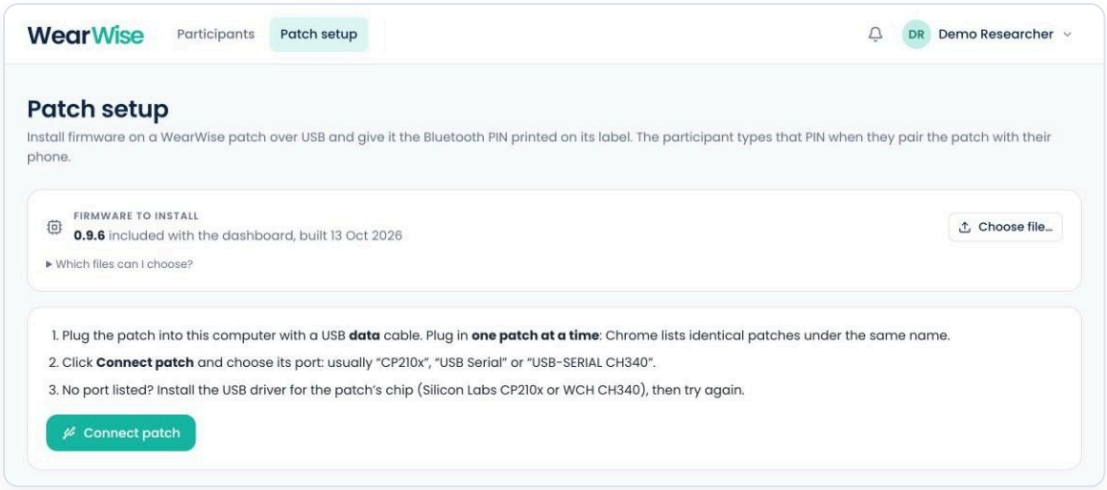
Alerts. Falls and readings that stayed out of range, for participants who share them with you.

Click the **bell** in the top bar to open **Alerts**. New alerts are marked New. *Status* shows how a fall was answered (*Marked OK* or unanswered) or whether an out-of-range reading has *Ended*. **View in session** opens the moment in its session.

- **Browser notifications:** click **Turn on** to get a notification when a new alert arrives while WearWise is open in that browser.
- **Alert sound:** a short tone for each new alert, and an urgent one for a possible fall. Browsers block sound until you've clicked the page once.
- **Refresh** reloads the list.

> **The dashboard is not a monitoring service**
>
> Alerts appear only while someone is looking at the dashboard. Nothing is sent by SMS or call, and nobody is on watch.

21 Patch setup and your account



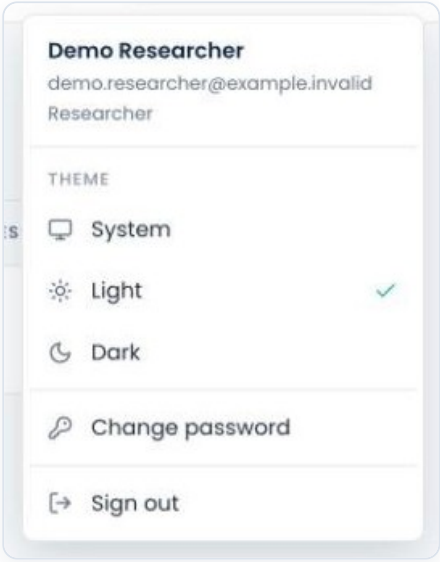
Patch setup installs firmware on a patch over USB and gives it the PIN printed on its label.

Patch setup (top bar) prepares a patch for a participant: plug it in with a USB data cable, click **Connect patch**, then choose **Set up** for a new patch, **Update** for new firmware, or **New PIN** for a new participant. Print the PIN on the patch's label. It only works in **Chrome or Edge**. The full steps are in the **WearWise Setup Guide**.

Your account menu

Click your name at the top right to see your role and:

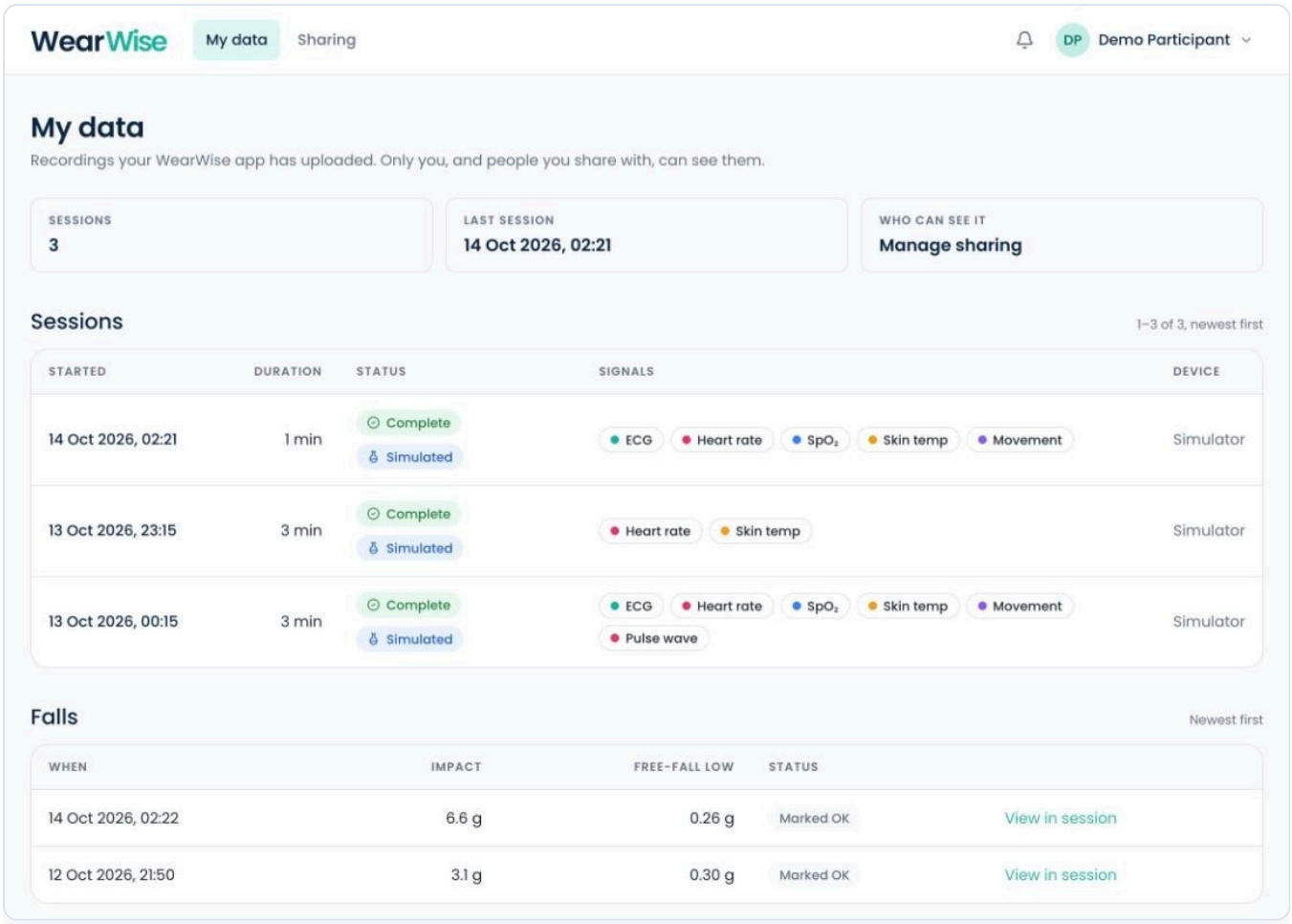
- **Theme:** **System**, **Light** or **Dark**.
- **Change password**: enter your current password and a new one twice. The checklist shows the rules.
- **Sign out**.



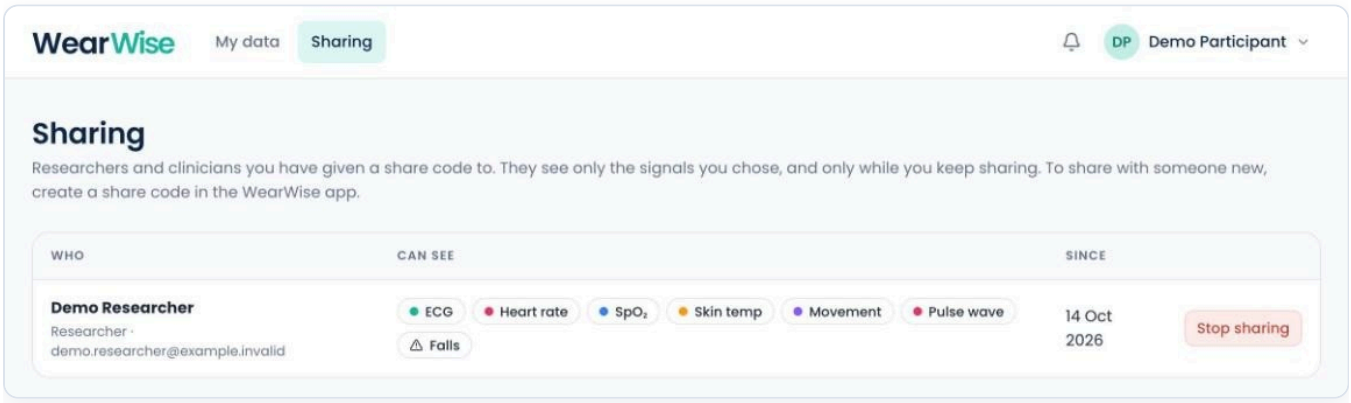
The account menu.

22 The dashboard for participants

Participants can sign in to the dashboard with their app account to see their data on a bigger screen. They get two pages:



My data. Your backed-up sessions and falls. Only you, and people you share with, can see them.



Sharing. Who can see your data and which signals. **Stop sharing** ends their access straight away. New share codes are made in the app.

PART 4

Help

Your privacy at a glance, and what to do when something doesn't work.

23 Your privacy at a glance

- ✓ Recordings are kept in an **encrypted database on your phone** and work fully offline.
- ✓ Finished sessions are backed up to the study's backend so they aren't lost with the phone.
- ✓ Health information you add to your profile **stays on your phone**.
- ✓ Nobody sees your data until **you** give them a share code. They see only the signals you chose, only while you keep sharing.
- ✓ Either side can end sharing at any time: **Remove access** in the app, or **Stop sharing** / **Remove participant** on the dashboard.
- ✓ Live view saves nothing and switches itself off.
- ✓ The patch never connects to the internet itself. Only your phone does.

24 Troubleshooting

Problem	What to do
The app says the backend isn't configured	This is a setup problem, not something you did. Tell the person who installed the app (Setup Guide, Troubleshooting).
No patch found when scanning	Check the patch is on and close by. Allow *Nearby devices* / Bluetooth. Close WearWise on any other phone: a patch talks to one phone at a time.
"The patch didn't accept this device"	Remove the old WearWise entry in the phone's Bluetooth settings, press **BOOT** on the patch, and pair again within 60 seconds, typing the label PIN.
A third phone or tablet can't pair	A patch remembers two devices. Press BOOT and pair: the oldest is replaced.
Heart rate about double at first, or SpO_2 around 80 %	Hold still for 10–20 seconds with steady, light finger pressure. Prototype modules can read poorly before settling.
ECG says "leads off"	An electrode isn't touching the skin. Check the electrodes and the lead colours.
The fall alarm keeps appearing with the simulation	That's on purpose (section 9). Tap **I'm OK**.
A session stays "not backed up"	Check your internet connection, then **Back up now** in Profile → Privacy & access.

Problem	What to do
The researcher can't see my sessions	Only backed-up sessions are shared. Also check that the code was used within 24 hours, and that you didn't make a newer code.
The live view button is greyed out	Connect your patch first.
The researcher sees nothing in live view	Live view is off, its time ran out, or WearWise is in the background (it pauses). Start it again and keep the app open.
Wi-Fi streaming never starts	Use a 2.4 GHz network. Bluetooth carries the data anyway.
The dashboard plays no alert sound	Click anywhere on the page once. Browsers block sound until you do.
Can't see Participants or Patch setup on the dashboard	Those pages are for researcher and clinician accounts. Ask the administrator.

25 Safety notes

Please read

- WearWise is a research prototype. It is not a medical device, and its readings are not clinically validated.
- "Possible fall detected" can be wrong, and real falls can be missed. Nobody is alerted outside WearWise.
- Never rely on WearWise in an emergency. Call emergency services directly.
- **ECG electrodes on a person:** the patch must run on a battery or USB power bank, never on a mains-powered computer or charger.
- Stop wearing the patch if it irritates your skin, gets hot, or is damaged.

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WearWise User Guide · handover release v1.0.0, October 2026. Screens show synthetic demo data. Security practices are informed by relevant guidance; nothing about WearWise is certified.